CHEMISTRY **ADAPTLY**

USING RESOURCES – THE HABER PROCESS

KEY VOCABULARY

Haber process: makes ammonia from nitrogen and hydrogen.

Reversible: the reaction reaches a dynamic equilibrium ( ).

Iron catalyst: speeds up the reaction without changing yield.

Compromise conditions: balance yield, rate and cost.

COMMON MISCONCEPTIONS

~~High temperature gives the best yield.~~

✓ The forward reaction is exothermic — high temp lowers yield.

~~Conditions are chosen for maximum yield.~~

✓ They are a compromise of yield, rate and cost.

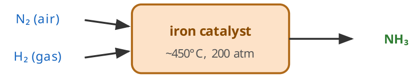
~~The catalyst increases the yield.~~

✓ It speeds up the reaction but does not change the yield.

HABER PROCESS

THE HABER PROCESS

$N_2 + 3H_2 \quad 2NH_3$



Unreacted N_2 and H_2 are recycled

Ammonia cooled and condensed off

1 THE EQUATION

Write the symbol equation for the Haber process. **[1 mark]**

2 RAW MATERIALS

State the sources of the nitrogen and hydrogen used. **[2 marks]**

3 TEMPERATURE CHOICE

Explain why about 450°C is used rather than a lower temperature. **[3 marks]**

Consider both rate and yield.

4 PRESSURE

Explain why a high pressure (200 atm) is used. **[2 marks]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

An _____ catalyst is used and unreacted gases are _____.

iron | recycled | copper | removed

6 USE OF AMMONIA

State the main use of the ammonia produced. **[1 mark]**

TIP: *$N_2 + 3H_2 \quad 2NH_3$. Conditions (~450°C, 200 atm, iron) are a compromise.*

✓ Think: Why is a moderate temperature a compromise between rate and yield?